

Abaid Ur Rehman

Machine Learning Engineer

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SUMMARY

Machine learning engineer specializing in computer vision, predictive modelling, and intelligent system automation. Proven ability to build and deploy end-to-end AI applications — from data preprocessing and model training to real-time Flask deployments. Experienced with deep learning (TensorFlow, YOLOv8), classical ML (XGBoost, Random Forest), and local LLM integration. Passionate about building practical, production-ready solutions that solve real-world problems.

PROFESSIONAL EXPERIENCE

Machine Learning & Artificial Intelligence

02/2026 – 05/2026

Techphantom, Faisalabad

- Completed hands-on training in supervised learning, computer vision, and NLP workflows.
- Built and evaluated classification and object detection models on real-world datasets.
- Gained practical experience deploying ML models via Flask-based web applications

EDUCATION

NFC Institute of Engineering and Fertilizer Research,
BS(Artificial Intelligence)

2023 – Present | Faisalabad

Kips College, Intermediate (Pre-Engineering)

2021 – 2023 | Faisalabad

Allied School, Matriculation (Science)

2019 – 2021 | Faisalabad

PROJECTS

Shop Management system

Built an end-to-end retail management solution featuring inventory control, product management, and automated billing to simplify daily business operations and enhance accuracy

Jarvis AI Desktop Assistant

An AI-powered desktop assistant capable of voice commands, application control, web search, file management, system automation, reminders, WhatsApp messaging, screenshots, and local AI conversations using Ollama.

Smart Waste Classification System

Smart Waste Classification System – AI-based web application using TensorFlow and Flask to classify waste into 6 categories with 91.5% accuracy. Features include real-time image prediction, live camera detection, top-3 results, recycling suggestions, analytics dashboard with Chart.js, voice feedback, and dark mode UI.

AI-Powered People Counting System

Engineered an intelligent people counting solution leveraging YOLOv8 and ByteTrack for real-time person detection, multi-object tracking, and occupancy monitoring. Integrated the system with a Flask-based dashboard to deliver crowd analytics and actionable insights.

Credit Card Fraud Detection System | Machine Learning, Flask

Developed an end-to-end machine learning fraud detection system using XGBoost, Random Forest, and Logistic Regression on 284,807 transactions. Addressed class imbalance with SMOTE, achieved a ROC-AUC score of 0.981, and deployed the solution as a Flask web application for batch fraud prediction via CSV/Excel uploads.

SKILLS

Languages

- Python
- HTML
- CSS
- JS

Data & Analytics

- Pandas
- NumPy
- Matplotlib
- Chart.js
- SMOTE (imbalanced data)

ML / DL Frameworks

- TensorFlow
- Keras
- scikit-learn
- XGBoost
- Random Forest
- YOLOv8

Web & Backend

- Flask
- REST APIs

NLP & AI Tools

- Ollama (local LLMs)
- NLTK
- Text classification

Tools & Platforms

- Git
- GitHub
- VS Code
- Jupyter Notebook
- Google Colab
- SupaBase

Computer Vision

- OpenCV
- ByteTrack
- real-time object detection & multi-object tracking